Q3. File-Based Systems vs DBMS

Q3

File-Based Systems vs DBMS

Comparison of traditional file-based systems and Database Management Systems (DBMS)

File-Based Systems

A traditional method of storing data in separate files managed by application programs.

VS

Database Management System (DBMS)

A software system that manages and controls data in a unified database with advanced features.

File-Based Systems	Aspect	Database Management System (DBMS)
Data stored in separate files (Flat files).	**Data Organization**	Data stored in an integrated and structured database.
No data independence. Programs depend on file structure.	**Data Independence**	Provides data independence. Changes in data do not affect programs.
High data redundancy (duplicate data in multiple files).	**Data Redundancy**	Minimizes data redundancy (single integrated database).
Low security. Easy to access and modify files.	**Security**	High security with access control and authentication.
Limited data sharing. Files are isolated.	**Data Sharing**	Supports multi-user access and data sharing.
Difficult to maintain and update data.	**Maintenance**	Easier maintenance with centralized control.
No backup and recovery mechanism.	**Backup & Recovery**	Provides backup and recovery mechanisms for data safety.

Advantages of DBMS over File-Based Systems

- Reduces data redundancy and inconsistency
- Provides data independence
- Ensures better security and access control
- Supports data sharing among multiple users
- Offers backup and recovery mechanisms
- Improves data integrity and consistency

In Summary

DBMS overcomes the limitations of file-based systems by providing a more efficient, secure and reliable way to store and manage data.

Key Idea: File-based systems are simple but have many limitations, while DBMS provides a structured, secure and efficient approach to data management.